

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I R. Kabi amudban (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

R. Kabi amudban
Signature of the student:

98
100
a

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 23	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

98

98
100

10

CO-1 Assessment Ques-1

Application Based Questions

3) (a) Demonstrate how the encryption process transforms the message:-

The keyword-based encryption can be demonstrated using the Vigenere Cipher.

Given:-

* Plain text :- HELLO

* keyword :- KEY

* Encryption rule :- $A=0, B=1, \dots, Z=25$

Since the plain text has 5 letters and the keyword has 3 letters, the keyword is repeated.

Position	1	2	3	4	5
Plaintext	H	E	L	L	O
Keyword	K	E	Y	K	E

So; plaintext :- HELLO
repeated keyword :- KEYKE

(i) Convert the letters to numbers:-

Letter	H	E	L	L	O
Plaintext value	7	4	11	11	14

Letter	K	E	Y	K	E
Key value	10	4	24	10	4

(ii) Apply the encryption formula.

$$C = (P + K) \bmod 26$$

$$1. H + K \Rightarrow (7 + 10) \bmod 26$$

$$= 17 \Rightarrow R$$

$$2. E + E \Rightarrow (4 + 4) \bmod 26 = 8$$

$$8 \Rightarrow I$$

$$3. L + Y \Rightarrow (11 + 24) \bmod 26 \Rightarrow 35 \bmod 26$$

$$\Rightarrow 9 \Rightarrow J$$

$$4. L + K \Rightarrow (11 + 10) \bmod 26 = 21$$

$$21 \Rightarrow V$$

$$5. O + E \Rightarrow (14 + 4) \bmod 26 = 18$$

$$18 \Rightarrow S$$

Final result:-

Cipher text :- RI JVS

(b) Explain how the repeating keyword influences each character of the Ciphertext.

The keyword is KEY, but the message HELLO contains five characters. Therefore, the keyword is repeated to match the message length :-

KEY $\rightarrow$ KEYKE

Each character of the plaintext is combined with the corresponding character of the repeated keyword.

Position	plaintext	key	Effect
1	H	K	H shifted using K $\rightarrow$ R
2	E	E	E shifted using E $\rightarrow$ I
3	L	Y	L shifted using Y $\rightarrow$ V
4	L	K	L shifted using K $\rightarrow$ V
5	O	E	O shifted using E $\rightarrow$ S

Thus; $H + K = R$, $E + E = I$, $L + Y = V$, $L + K = V$, $O + E = S$
The keyword KEY therefore controls the shift applied to each other character. When the plaintext is longer than the keyword, the keyword repeats.

This produces different shifts for different characters rather than applying one fixed shift to the entire message. Hence, the resulting cipher text is RIJVS.

Conclusion:- The plaintext HELLO, encrypted using the repeating keyword KEY, produces RIJVS. The repeated keyword determines the individual shift applied to each plaintext character, providing a polyalphabetic substitution mechanism.

5. Legacy Encryption in Corporate communication.

(a) Determine the most probable original message appears as, by reversing the encryption process. the given cipher text is:

WECRLTEERDSOEFEAOCAIVDEN

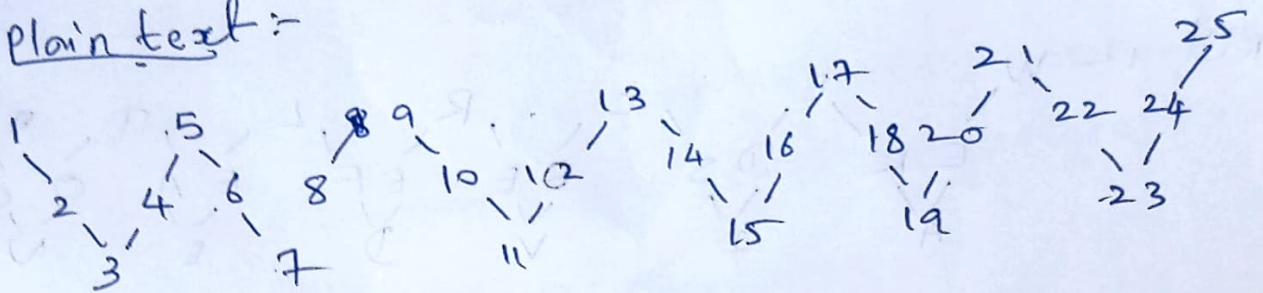
The encryption method uses three levels of arrangement, which can be treated as a 3-rail transposition arrangement.

(i) The cipher text contains 25 characters. For Three rails, the zigzag pattern is:-

Rail₁ → Rail₂ → Rail₃ → Rail₂ → Rail₁ → ...

The pattern looks like :-

Plain text :-



Therefore, the 25 characters are distributed as :-

* Rail 1 = 7 characters

* Rail 2 = 12 characters

* Rail 3 = 6 characters

$$7 + 12 + 6 = 25$$

(ii) Divide the ciphertext into the three rails

Given ciphertext :-

WE CRLTEERD SOEEFEAOCAI VDEN

Take first 7 characters for Rail 1 :-

WE CRL TE

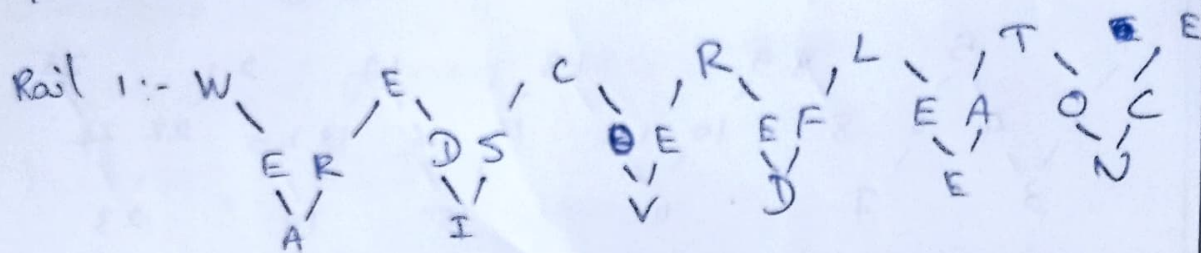
Take the next 12 characters for Rail 2 :-

ERD SOEEFEAO C

Take remaining 6 characters from Rail 3 :-

AI VDEN

(iii) place the characters back into the zigzag pattern



Now let's read the ~~read~~ characters following the zigzag path:-

WEAREDISCOVEREDFLEEATONCE

WE ARE DISCOVERED FLEE AT ONCE

(b) Critically evaluate why such a method is vulnerable in modern communication systems. The Rail Fence cipher is a transposition cipher. It does not replace the letter; instead, it changes their positions. Therefore, the original letters and their frequencies remain unchanged.

1. Small key space:-

The main key is the number of rails. For a simple rail Fence cipher, an attacker can try different numbers of rails and check which result produces meaning full text. So, the number of possible keys is very small compared with modern cryptographic systems.

2. Letter frequencies are preserved:-

The cipher only rearranges characters.

For example, if the plaintext contains a particular number of E characters, the ciphertext will contain the same number of E characters.

Therefore, statistical information about the plaintext is not hidden.

3. Vulnerable to brute-force analysis

Because there are relatively few possible rail configurations, an attacker can test different rail counts until a meaningful English message appears. For a short message such as this one, this is especially easy.

4. Does not provide strong confidentiality.

The encryption does not use a strong mathematical transformation or a large cryptographic key. It simply rearranges the positions of the characters. Therefore, it cannot provide the level of confidentiality required for modern corporate communication.

5. No authentication or integrity protection

A basic rail fence cipher only provides scrambling of the message.

It does not provide a mechanism to verify:-

- * who sent the message.
- * whether the message was modified, or
- * whether the ciphertext is authentic.

6. Unsuitable for modern security

Because of its small key space and simple transposition mechanisms, Rail Fence is mainly useful for learning classical cryptography, not for protecting sensitive modern communication.

Therefore, Rail fence Cipher = Simple classical cipher, not modern secure encryption.

c) Propose one practical enhancement to improve the confidentiality of messages.

A practical enhancement is to replace the legacy transposition cipher with AES-based encryption.

Proposed solution:-

Legacy Transposition Cipher $\rightarrow$ AES.

AES (Advanced Encryption Standard) is a modern Symmetric encryption algorithm and provides Substantially stronger confidentiality than a simple transposition cipher.

A more appropriate modern implementation would be AES-GCM, because it provides both encryption and protection against message modification.

Improved Communication process :-

